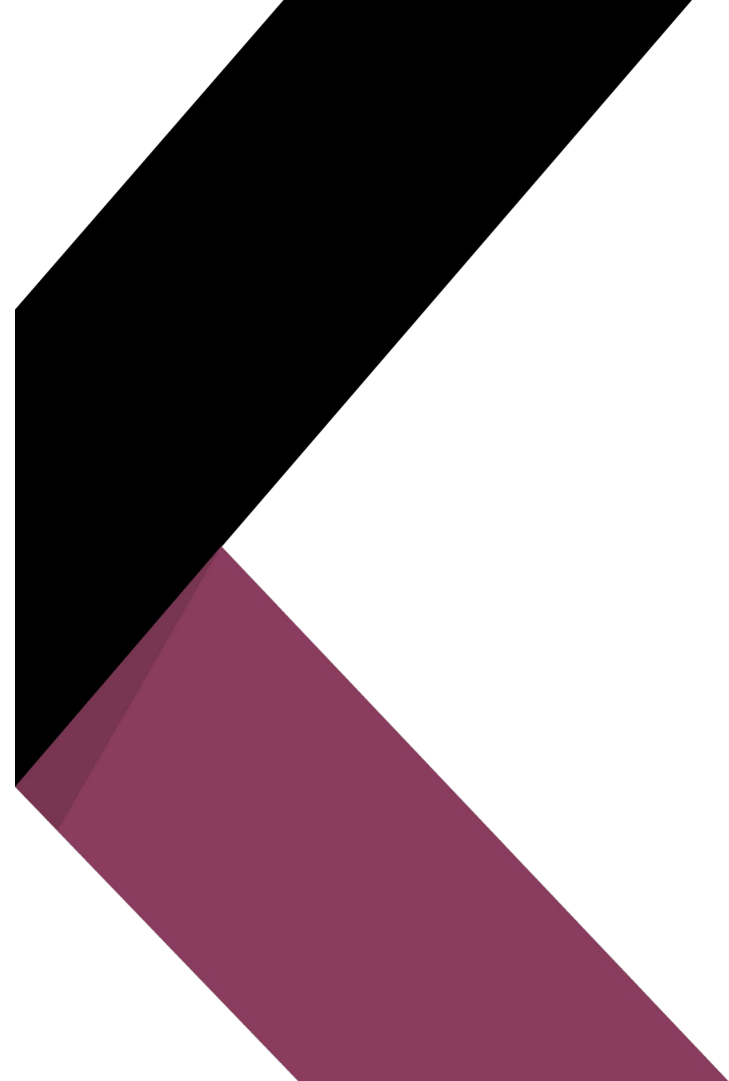


Test Automation

NI HIL





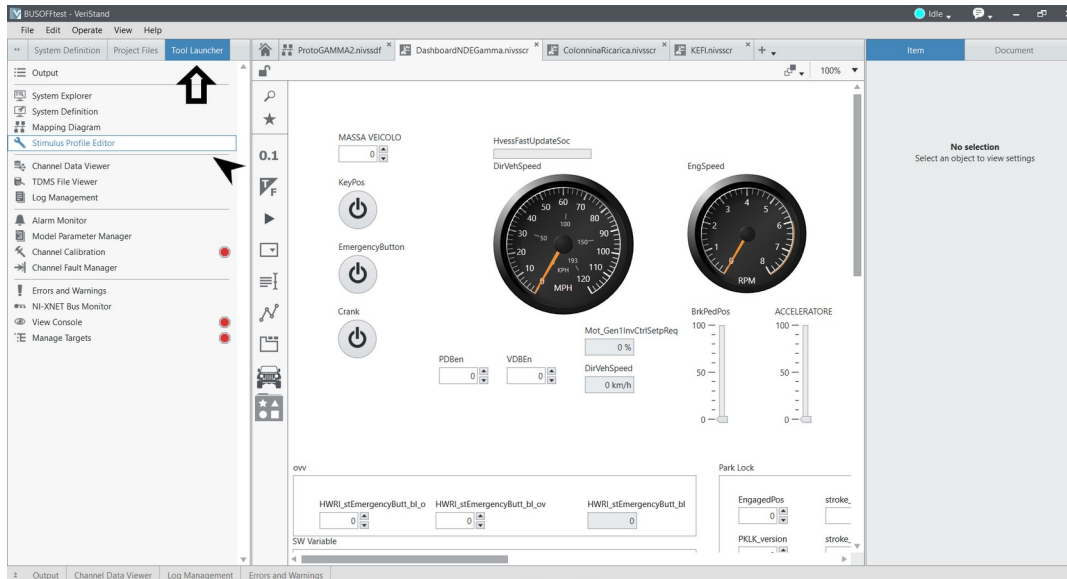
Definition of Sequences

- The term "Test Automation" refers to the process of reviewing and validating a software carried out in a totally automatic manner, exempting the tester from manual interventions. This strategy allows a substantial reduction in the time required for testing and re-testing activities. Test Automation also allows you to dispose of the bugtracking phase thanks to the process output, called a report, in which not only the result of the test is highlighted but above all, in case of failure, the cause for which the validation has given a negative result. In order to carry out tests automatically, it is necessary to define a UUT (Unit Under Test), in our case the control unit, and a coded structure that allows automatic interaction with the UUT, defined as a Framework, which also includes the definition of the VeriStand project on which we are working.

Stimulus Profile Editor

Framework

- The software that was used to write the code is an NI VeriStand tool called "Stimulus Profile Editor". This interface is similar to the more standard programming tools (such as CodeBlocks or Xcode) but allows you to manipulate parameters defined in the current VeriStand project

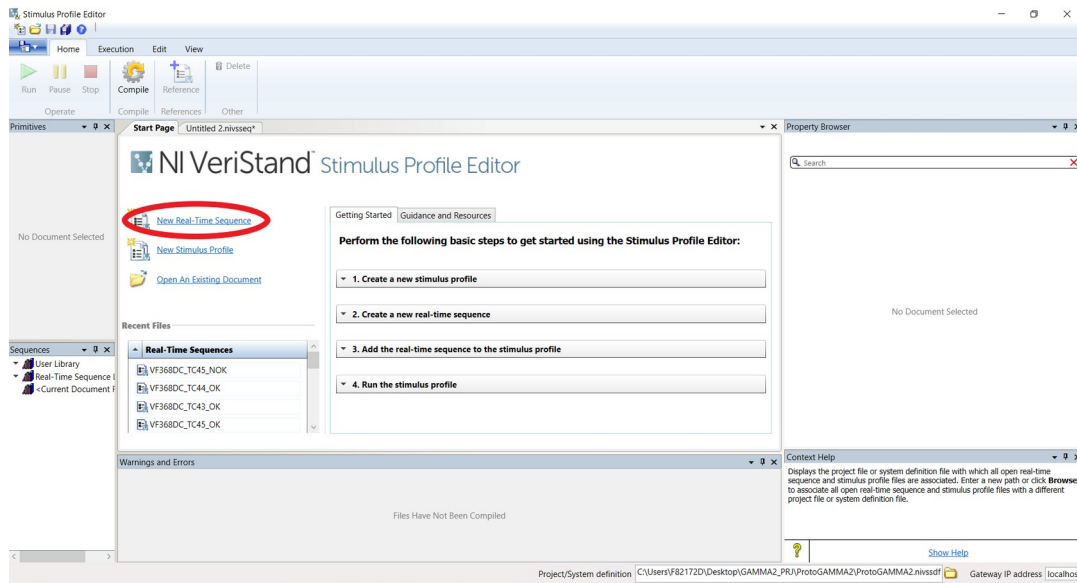


After starting VeriStand and a previous project, click on the "Tool Launcher" and "Stimulus Profile Editor" window to start the tool

Libraries – Stimulus Profile Editor

Framework

- The tool screen has a dedicated menu on the "Start Page"



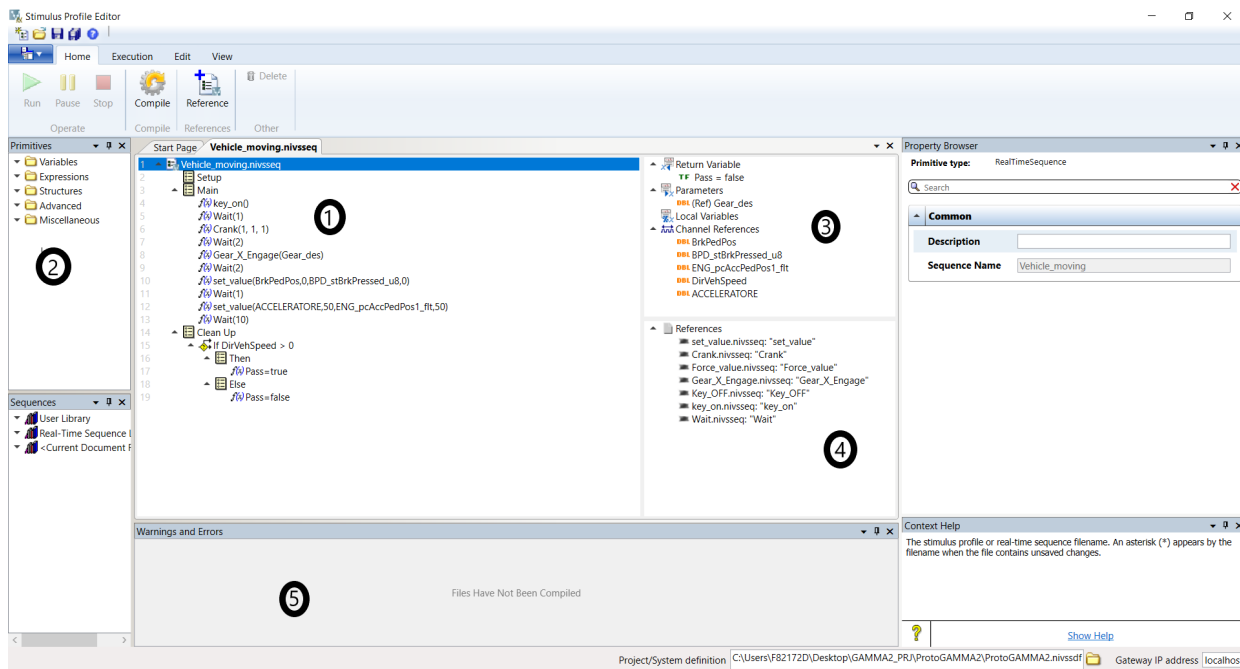
The Real-Time Sequence is used to encode the maneuvers performed in a single test, so they will represent the translation into code of a single test case

- In order to begin to turn the maneuvers to be carried out during the test into code, the "structure" "Real-Time Sequence" will be used.

Librerie – Stimulus Profile Editor

Framework

- The often recurrence of "type" maneuvers has made it necessary to create "libraries" to allow their recall during test cases: an example is the sequence of maneuvers that allows the emulation of the moving vehicle. The following example is proposed:



The syntax used to implement the libraries is absolutely analogous to that of test cases

- As you can see in the previous slide, the interface contains several sections:
 - sector '1' contains code syntax
 - The reference "2" is in the "Primitives" section which contains (a) the folder of the various types of "Variables" possible to be declared, (b) the folder of "Expressions" or assignments, through which to assign values to parameters/variables or references, (c) the folder of the structures that contains the loops that can be used (while, do while, for cycles...), the conditionals (if/else) or the "Multitasking" in case a simultaneous management of several instructions is required (Management of multiple tasks such as executing several instructions at the same time, stopping one before the others, and so on), (d) the "Advanced" folder that allows you to invoke a state for suspending the sequence in real time while the primary control loop iterates in real time, (e) the "Miscellaneous" folder that allows you to document the code or create a list of states
 - It is the panel of the quantities involved/recalled in the sequence, in particular: (a) The "Return Variable", i.e. the output indicator of the sequence, which can be a Boolean (in case you check whether the test is OK or KO) or a double (real numerical value), (b) the "Parameters", i.e. the input variables of the sequence / function, (c) The "Local Variables", from the name local variables defined within the sequence, (d) the "Channel References", i.e. a link to the parameters defined internally by the project (Aliases, CAN signals, software variables, analog and/or digital signals)
 - The "References" section contains a list of all the real-time sequences referred to in the selected real sequence
 - Finally, the Panel that indicates errors or warnings following compilation

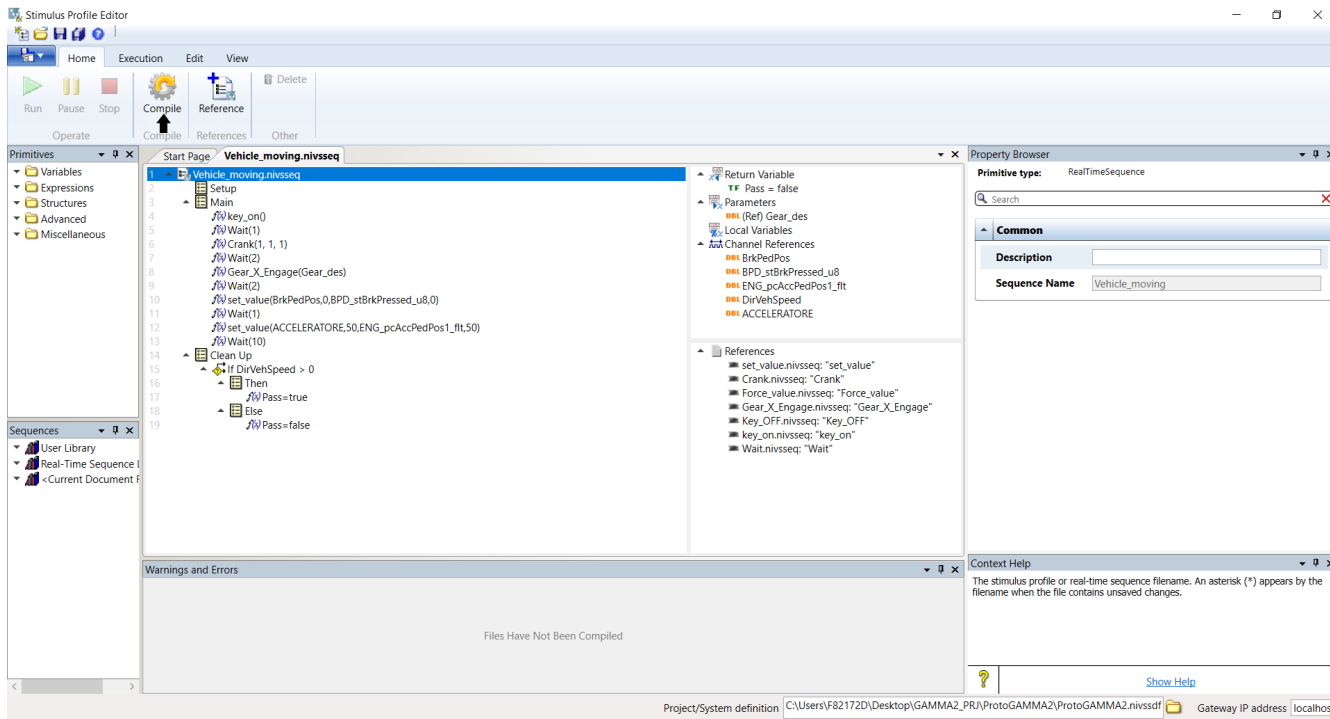


Real-Time Sequences Running

Compiling the Real-Time Sequence

- Once defined, we proceed with the compilation

Compile

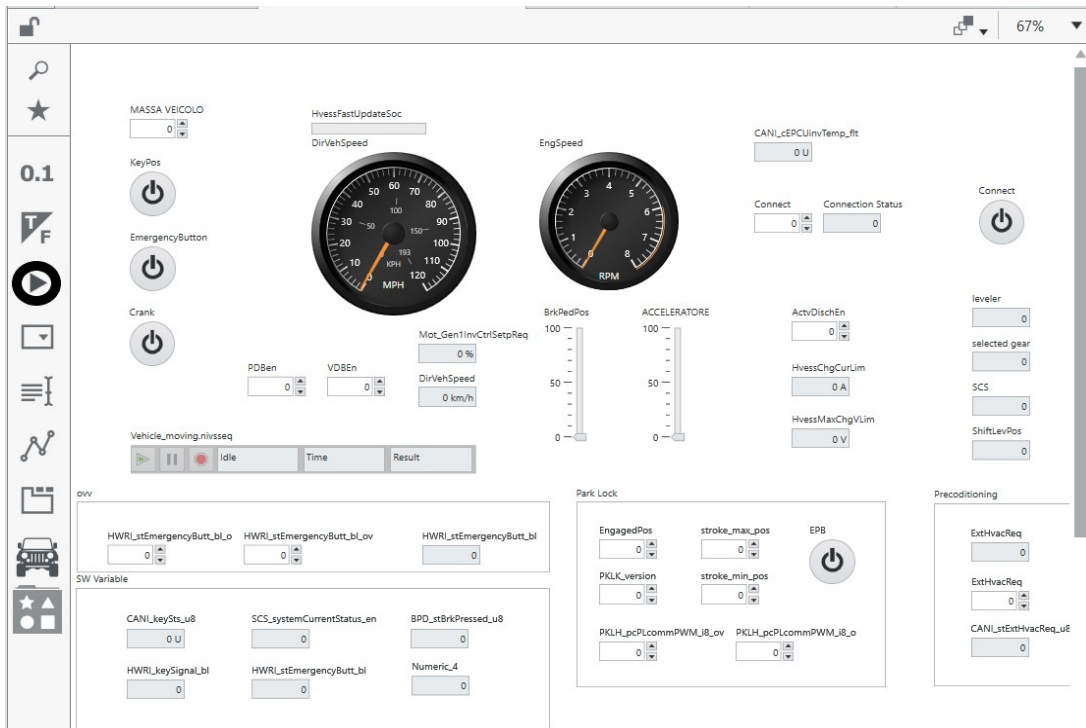


If, in the case like this, the compilation will not give errors or warnings, we proceed with the launch

Real-Time Sequence Running

Run

- At this point the project and the dedicated screen open, containing the parameters of projects involved in the sequence: in the case in question the screen is a simplified representation of the vehicle-driver interface.



Opening the selected icon in the menu to left, click on the third option for invoke a control of Real-Time Sequence.

By pressing the "Play" button, the sequence is launched and, once finished, the result will take the value of the Return Variable (in the case of false boolean if the test is KO or true in test case OK, while the value of the output if you are using a double)

If you record consistency between the manual test result and that of the sequence, then the test case was automated properly



VF Automation

Stimulus Profile

Automation

The screenshot displays the Stimulus Profile Editor application. The interface includes a menu bar (Home, Execution, Edit, View), a toolbar with icons for Run, Pause, Stop, Compile, Reference, and Delete, and a main workspace. On the left, a 'Steps' tree view shows a hierarchy of actions, with 'Project' expanded and 'Open VeriStand Project' selected. Two black arrows point to 'Real-Time Sequence Call' and 'Project' in this tree. The main workspace shows a sequence of steps: 1. Setup, 2. Real-Time Sequence: [Target: <Auto>], 3. Main, 4. Clean Up. Below the workspace is an 'Execution Results' section. At the bottom, a 'Warnings and Errors' pane shows 'No Errors or Warnings'. On the right, a 'Property Browser' for 'RealTimeSequenceCallStep' is visible, showing fields for Description, File Path, Target Name, and Timeout (ms), along with a 'Pass/Fail Evaluation' section and an 'Update Parameters' button. A 'Context Help' pane at the bottom right provides information about the Compile button's function.

Stimulus Profile Editor

Home Execution Edit View

Run Pause Stop Compile Reference Delete

Operate Compile References Other

Steps

- Real-Time Sequences
 - Real-Time Sequence Call
 - Real-Time Sequence Group
- Logging
 - Start Logging
 - Channel Group
 - Stop Logging
- VeriStand Control
- Project
 - Open VeriStand Project
 - Close Active VeriStand Project
 - Control Active VeriStand Project
- Workspace
 - Workspace Tools
 - Open VeriStand Workspace
 - Close VeriStand Workspace
 - Update Model Parameters from File
 - Launch NI VeriStand
- Other

Start Page Vehicle_moving.nivseq Untitled 1.nivstimprof*

1. Setup

2. Real-Time Sequence: [Target: <Auto>]

3. Main

4. Clean Up

Execution Results

Warnings and Errors

No Errors or Warnings

Property Browser

Step type: RealTimeSequenceCallStep

Search

Common

Description

File Path Select a path

Target Name

Timeout [ms] 10000

Pass/Fail Evaluation

Type AlwaysPass

Parameters No Parameters

Parameters

Update Parameters

Context Help

When you click **Compile**, the **Warnings and Errors** pane lists any errors, warnings, or messages in a stimulus profile or real-time sequence file. Click an error, warning, or message in the **Warnings and Errors** pane to display a description and, if applicable, a suggested solution for the error or warning.

Show Help

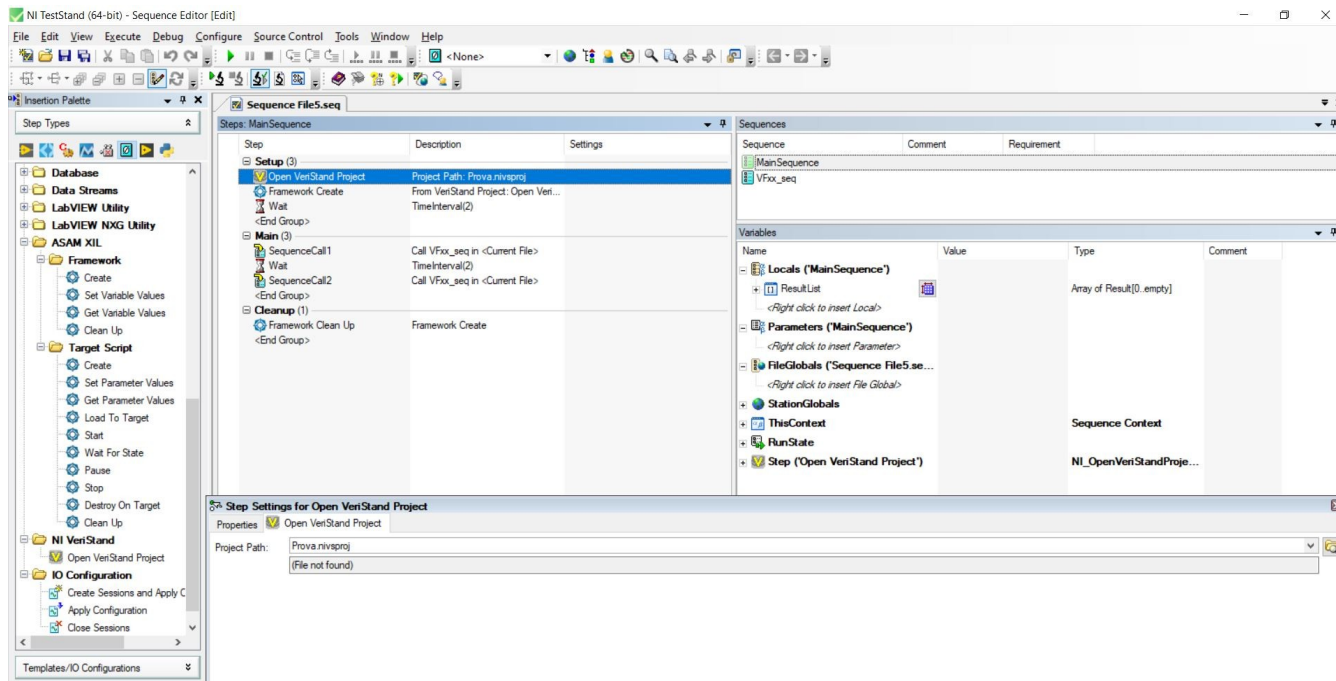
Project/System definition C:\Users\F82172D\Desktop\GAMMA2_PRJ\ProtoGAMMA2\ProtoGAMMA2.nivssdf Gateway IP address localhost

- The priority operation is the recall of the "VeriStand Project", through the path of the folder on the left "VeriStand Control", thus being able to open or close the project and actively control it. We then move on to the recall of the various Real-Time Sequences, which represent the individual test cases of the VF, through the command "Real-Time Sequence Call" and in which, as observed in the previous slide, it is necessary to define the "File Path" of the sequence, the "Target Name", or the identification of the target with all its specifications, such as "the Operating System", "IP Address (address)", the tasks of the processor, the "Data Processing Loop" and other settings such as the processor execution mode (Low Latency or Parallel), the frequency rate of the DAQ and the DIOs, the "warmup time" and finally, perhaps the most significant, the "Target rate (Hz)" and the "Timing Source Timeout (ms)". Finally, the evaluation of the Pass/Fail is defined which, as in the case of the single Real-Time Sequence, can consist of "Boolean", "Always Pass" or a "NumericBoundsCheck" output in the case of control against a predefined threshold. Finally, the "Update Parameters" function that allows you to modify the input parameters to study the reaction of the UUT to the test. Once the writing is complete, we proceed to the compilation and the subsequent launch. At the end of the "Run", a sort of Report will be generated that simply indicates the result of each Real-Time Sequence.

- The second possibility of automatic execution of the Test Cases of an entire Vehicle Function deals with the use of the TestStand Tool, a separate sw compared to VeriStand, but still a product of National Instruments. In our specific case TestStand will not be used to write sequences in Phyton, C++ or LabView, but to invoke the Real-Time Sequences already written on Stimulus Profile Editor, thus using it instead of a draft of a Stimulus Profile. The main screen is proposed in the next slide, together with the basic steps to follow for the automation sequence. The screen is presented with the "Main Sequence" in which the various "Real-Time Sequences" will be called up via the "Sequence Call" function in the central part of the code.

NI TestStand – Main Sequence: open VeriStand Project

Automation



1. The first step is to perform the preliminary operations in the Setup Section in the Main Sequence, specifically:
1. Project Declaration and Project Opening by selecting the path on the left "NI VeriStand"  "Open VeriStand Project"
In the path on the left, once you have selected "ASAM XIL", select the Framework item since you have to create the Framework based on the open VeriStand project, as you can see
In the next slide

NI TestStand – Main Sequence: Creating the Framework

Automation

The screenshot displays the NI TestStand Sequence Editor interface. The main workspace shows a sequence named 'MainSequence' with the following steps:

- Setup (3)**
 - Open VeriStand Project (Project Path: Prova.nviproj)
 - Framework Create (From VeriStand Project, Open VeriStand Project)
 - Wait (TimeInterval(2))
- Main (3)**
 - SequenceCall1 (Call VFxx_seq in <Current File>)
 - Wait (TimeInterval(2))
 - SequenceCall2 (Call VFxx_seq in <Current File>)
- Cleanup (1)**
 - Framework Clean Up

The 'Framework Create' step is selected, and its properties are shown in the 'Step Settings for Framework Create' panel:

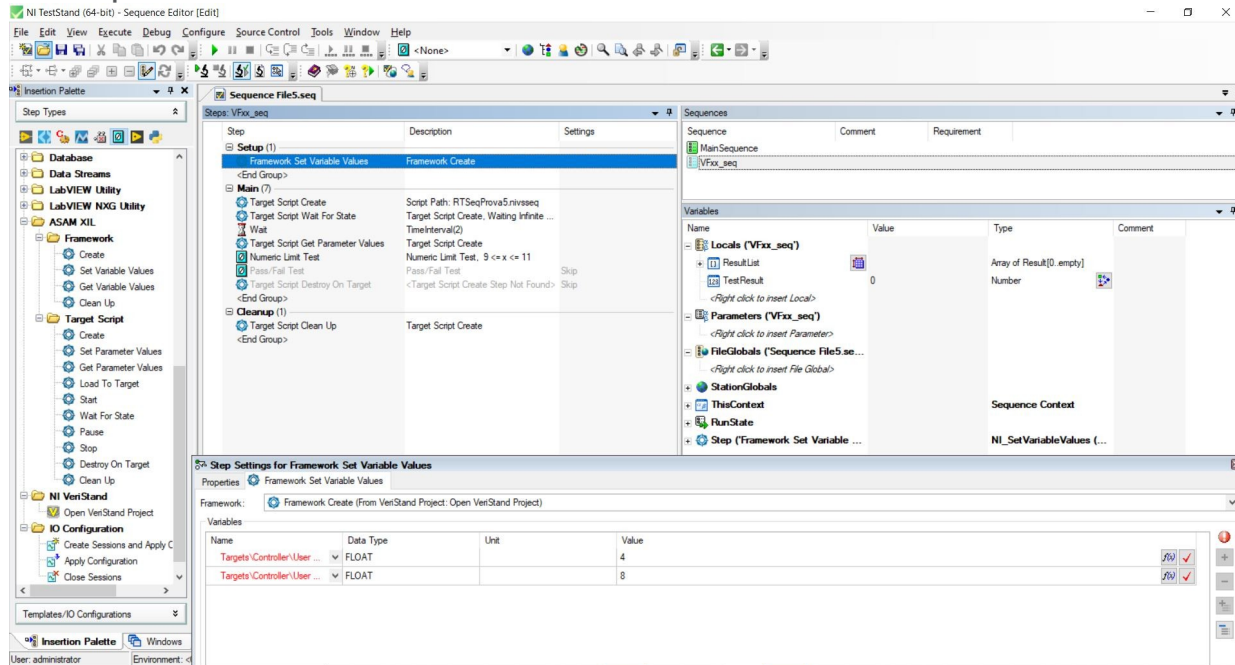
- Properties:** Framework Create
- From Manifest:** ☐
- From VeriStand Project:** ☒ Open VeriStand Project (Project Path: Prova.nviproj)

The 'Sequences' pane on the right shows the 'MainSequence' and 'VFxx_seq' sequences. The 'Variables' pane shows the 'Locals (MainSequence)' section with a 'ResultList' variable of type 'Array of Result[0..empty]'. The 'Parameters (MainSequence)' section is also visible.

NI TestStand – VFxx_seq: Sequence-Interface

Automation

- 2. We proceed to the definition of a sort of sequence-bridge, which allows the recall of the Real-Time Sequence defined on Stimulus and the management of its parameters



(a) Declaration of the 'Test Result' among the variables

(b) In the Setup section, please call the command "Set variable values" which Allows the set of parameters defined as «User Channels» in the System Definition File of the VeriStand Project (as seen in the path name)

(c) You go to the "Main" section of the sequence, in which to create the Target script by calling it from the left menu, in which to define the path of the project and the Real-Time Sequence file created on Stimulus Profile Editor

NI TestStand – VFxx_seq: Sequence-Interface

Automation

(d) Invoke the command "Target Script Wait For State" to align the target with the "Finished Run" condition of the RTS, and then put a Wait that has a synchronization function with the target (time buffering)

(e) Invoke the command "Target Script Get Parameter Values" to read the local variable "Test Result"

The image displays two screenshots from the NI TestStand software interface. The top screenshot shows a sequence editor with a tree view on the left and a table of sequence steps on the right. The tree view includes steps like 'Target Script Wait For State', 'Wait', 'Target Script Get Parameter Values', 'Numeric Limit Test', 'Pass/Fail Test', 'Target Script Destroy On Target', and a 'Cleanup' group. The table on the right lists the details for each step, including the script name and any associated conditions or timeouts. The bottom screenshot shows the 'Step Settings for Target Script Get Parameter Values' dialog box. It has a 'Properties' tab and a 'Script' section where 'Target Script Create (Script Path: RTSeqProva5.nvseq)' is selected. Below this, there are checkboxes for 'Wait For States' (In Configuration, Ready, Running, Paused, Finished, Stopped), with 'Finished' being checked. At the bottom, there is a 'Parameters' section with a table for defining parameters.

Name	Parameter Type	Data Type	Value
Pass	OUTPUT	FLOAT	Locals.TestResult

(f) At this point you use the "Action Test" command which allows you to select the test mode (Limit Test Numbers or a Pass/Fail test)

(g) Finish writing the sequence defining, in the CleanUp section, the "Target Script Clean Up" to clear the target script created for the current sequence and return to initial conditions

NI TestStand – Main Sequence

Automation

- 3. The VF automation process is completed by completing the definition of the Main Sequence, in particular by recalling, in cascade, through the "Sequence Call" command, the VFxx sequences each of which must contain a single RTS, i.e. a single test case. There will be many sequence calls in the Main Sequence, there will be many test cases of the VF in question.

The screenshot displays the NI TestStand (64-bit) - Sequence Editor [Edit] window. The main workspace shows the 'MainSequence' sequence with the following steps:

- Setup (3)**
 - Open VeriStand Project
 - Framework Create
 - Wait
 - <End Group>
- Main (3)**
 - SequenceCall1
 - Wait
 - SequenceCall2
 - <End Group>
- Cleanup (1)**
 - Framework Clean Up
 - <End Group>

The 'SequenceCall1' step is selected, and its 'Step Settings for SequenceCall 1' dialog is open. The settings are as follows:

Property	Value
File Pathname:	<Current Sequence File>
File Pathname:	C:\Users\F82172D\Downloads\Sequence File5.seq
Sequence:	VFxx_seq
Sequence Comment:	

The 'Execution Options' section shows:

- ☐ Specify By Expression
- ☒ Use Current File
- Execution Options: None
- ☒ Use Prototype of Sequence

The 'Variables' pane on the right lists the following variables:

- Locals ('MainSequence')**
 - ResultList: Array of Result[0..empty]
- Parameters ('MainSequence')**
 - <Right click to insert Parameter>
- FileGlobals ('Sequence File5.se...')**
 - <Right click to insert File Global>
- StationGlobals**
 - ThisContext: Sequence Context
- RunState**
 - RunState: SequenceCall (Step)

By compiling the sequence, you get the output is the full report, can also be exported as a PDF file

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